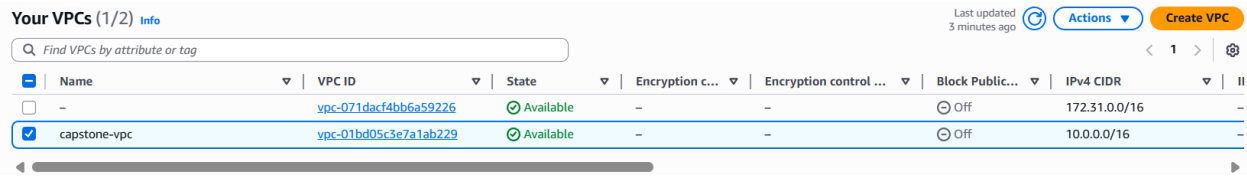
Capstone Project: Enterprise AWS Infrastructure Deployment

2. Networking Layer (VPC Design)

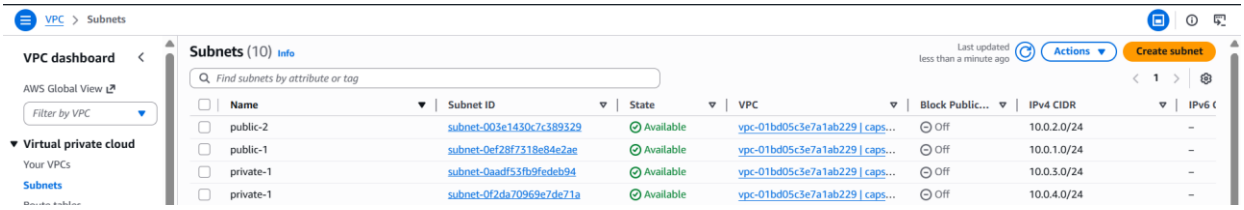
Step 1 Create a VPC : capston-vpc



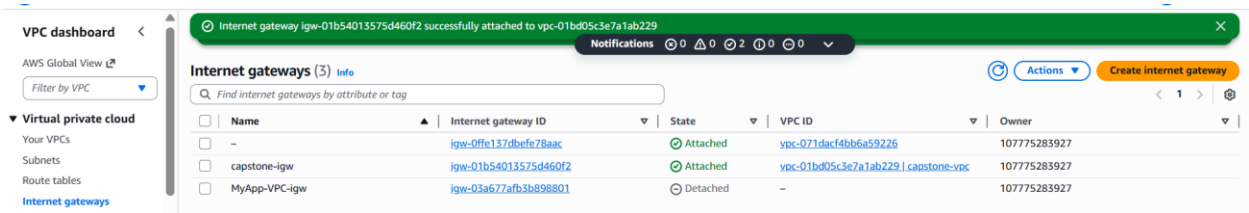
Step 2 Create 4 subnets:

2 Public Subnets (different Availability Zones)

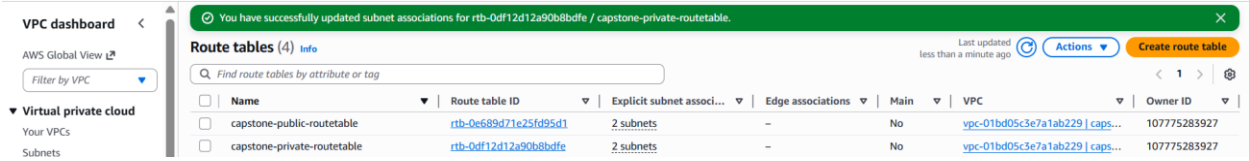
2 Private Subnets (different Availability Zones)



Step 3 create internet gateway: capston-igw and attach with the VPC



Step 4 create 2 route tables and associate with subnets:



Route tables (1/4) Info

Last updated 3 minutes ago | Actions | Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
capstone-public-routetable	rtb-0e689d71e25fd95d1	2 subnets	–	No	vpc-01bd05c3e7a1ab229 | caps...	10777528...
capstone-private-routetable	rtb-0df12d12a90b8bdfe	2 subnets	–	No	vpc-01bd05c3e7a1ab229 | caps...	10777528...
–	rtb-0436fb47b72b9f40f	–	–	Yes	vpc-071dacf4bb6a59226	10777528...
–	rtb-0b2517df1e066fb9f	–	–	Yes	vpc-01bd05c3e7a1ab229 | caps...	10777528...

rtb-0e689d71e25fd95d1 / capstone-public-routetable

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Routes (2)

Both | Edit routes

Filter routes

Destination	Target	Status	Propagated	Route Origin
0.0.0.0/0	igw-01b54013575d460f2	Active	No	Create Route
10.0.0.0/16	local	Active	No	Create Route Table

Route tables (1/4) Info

Last updated 3 minutes ago | Actions | Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
capstone-public-routetable	rtb-0e689d71e25fd95d1	2 subnets	–	No	vpc-01bd05c3e7a1ab229 | caps...	10777528...
capstone-private-routetable	rtb-0df12d12a90b8bdfe	2 subnets	–	No	vpc-01bd05c3e7a1ab229 | caps...	10777528...
–	rtb-0436fb47b72b9f40f	–	–	Yes	vpc-071dacf4bb6a59226	10777528...
–	rtb-0b2517df1e066fb9f	–	–	Yes	vpc-01bd05c3e7a1ab229 | caps...	10777528...

rtb-0e689d71e25fd95d1 / capstone-public-routetable

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
public-1	subnet-0ef28f7318e84e2ae	10.0.1.0/24	–
public-2	subnet-003e1430c7c389329	10.0.2.0/24	–

Route tables (1/4) Info

Last updated 4 minutes ago | Actions | Create route table

Find route tables by attribute or tag

Name	Route table ID	Explicit subnet associ...	Edge associations	Main	VPC	Owner ID
capstone-public-routetable	rtb-0e689d71e25fd95d1	2 subnets	–	No	vpc-01bd05c3e7a1ab229 | caps...	10777528...
capstone-private-routetable	rtb-0df12d12a90b8bdfe	2 subnets	–	No	vpc-01bd05c3e7a1ab229 | caps...	10777528...
–	rtb-0436fb47b72b9f40f	–	–	Yes	vpc-071dacf4bb6a59226	10777528...
–	rtb-0b2517df1e066fb9f	–	–	Yes	vpc-01bd05c3e7a1ab229 | caps...	10777528...

rtb-0df12d12a90b8bdfe / capstone-private-routetable

Details | Routes | Subnet associations | Edge associations | Route propagation | Tags

Explicit subnet associations (2)

Edit subnet associations

Find subnet association

Name	Subnet ID	IPv4 CIDR	IPv6 CIDR
private-1	subnet-0aadf53fb9fedeb94	10.0.3.0/24	–
private-1	subnet-0f2da70969e7de71a	10.0.4.0/24	–

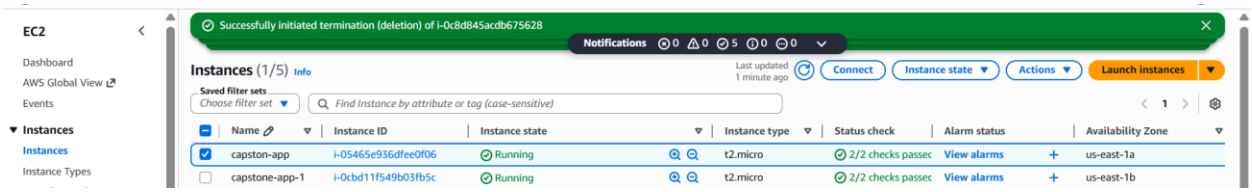
Step 5 create Nat Gateway:

NAT Gateway was designed but not deployed due to sandbox IAM restrictions. Private subnet design remains part of architecture.

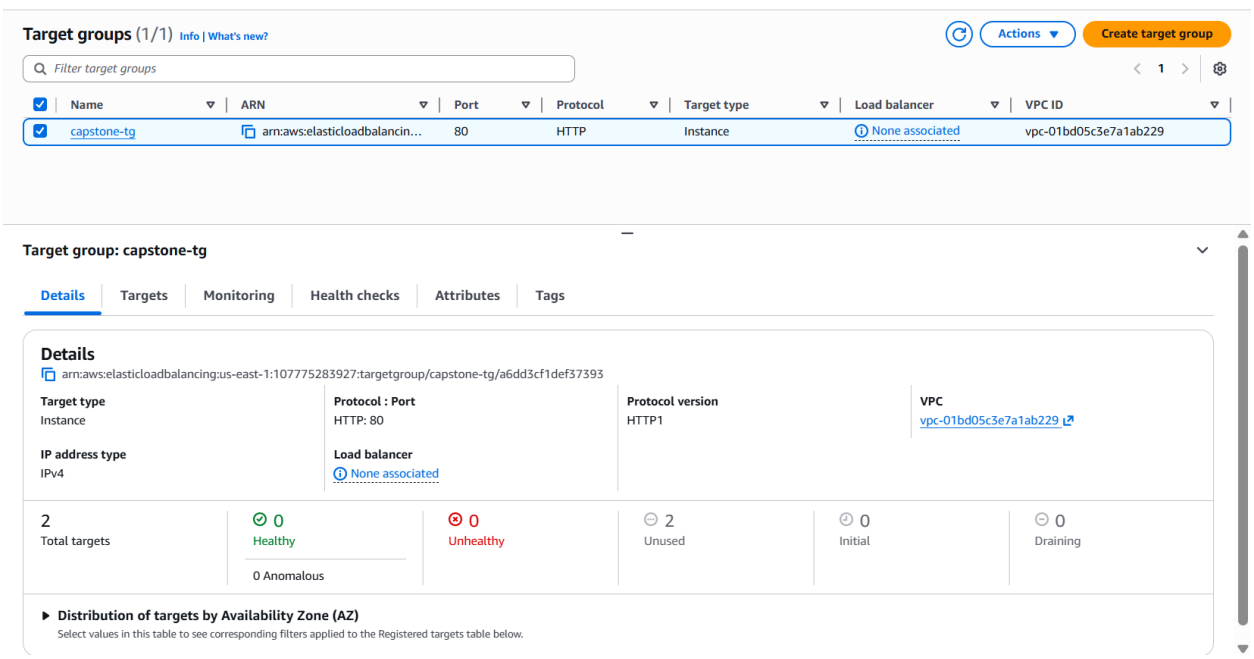
3. Compute Layer

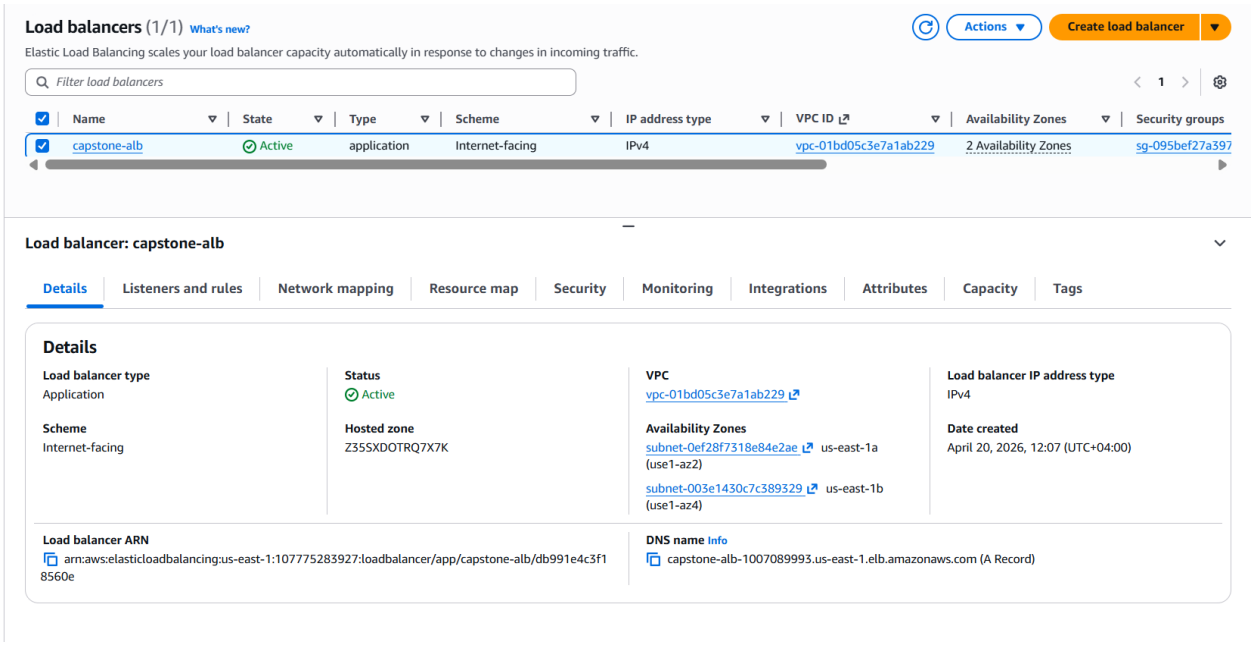
created 2 EC2 instances, Nginx installed thro user data and Instances placed in private subnets.
user data:

```
#!/bin/bash
yum update -y
yum install -y nginx
systemctl start nginx
systemctl enable nginx
echo "<h1>Capstone App - Instance 2</h1>" > /usr/share/nginx/html/index.html
```

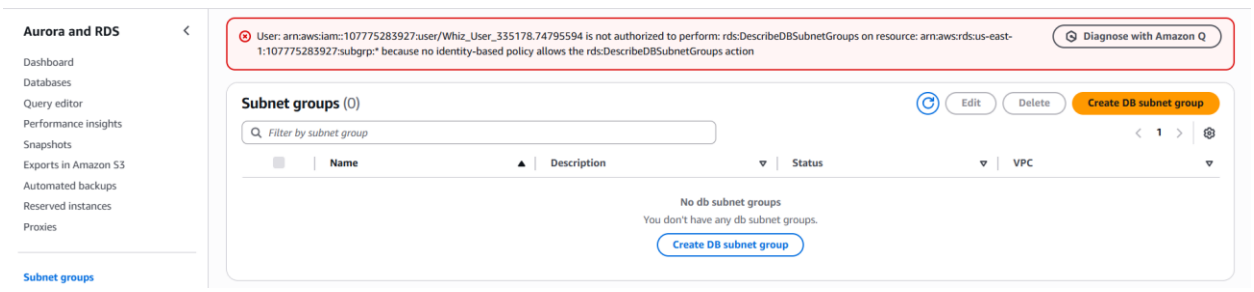


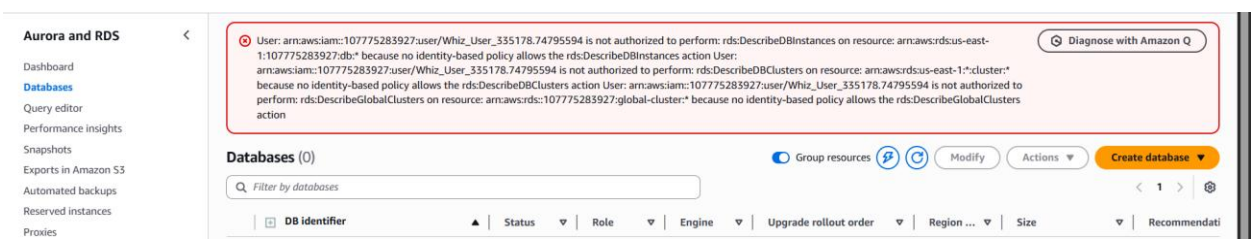
4. Load Balancing





5. Data Layer



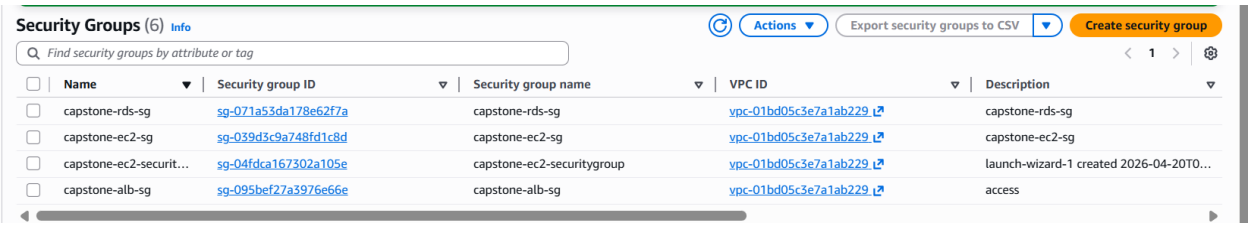


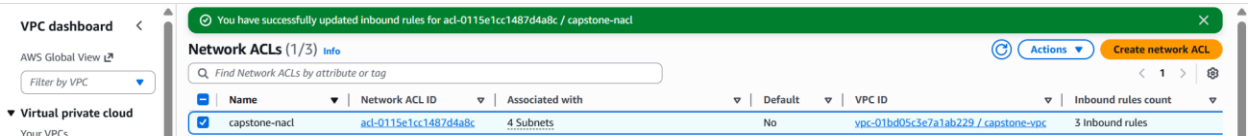
The Data Layer (Amazon RDS MySQL) was designed to be deployed in private subnets with access restricted only to the application layer (EC2 instances) via security groups. However, deployment of DB Subnet Group and RDS was not possible due to sandbox IAM restrictions prevent creation of RDS networking resources.

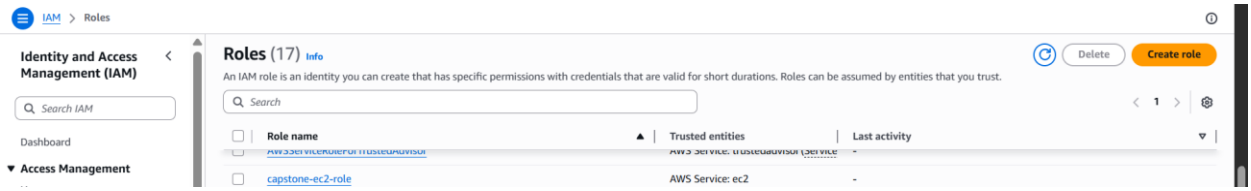
The architecture design still includes:

- Private DB subnets
- DB subnet group (design-level)
- Security group restricting access to EC2 only
- No public access to database

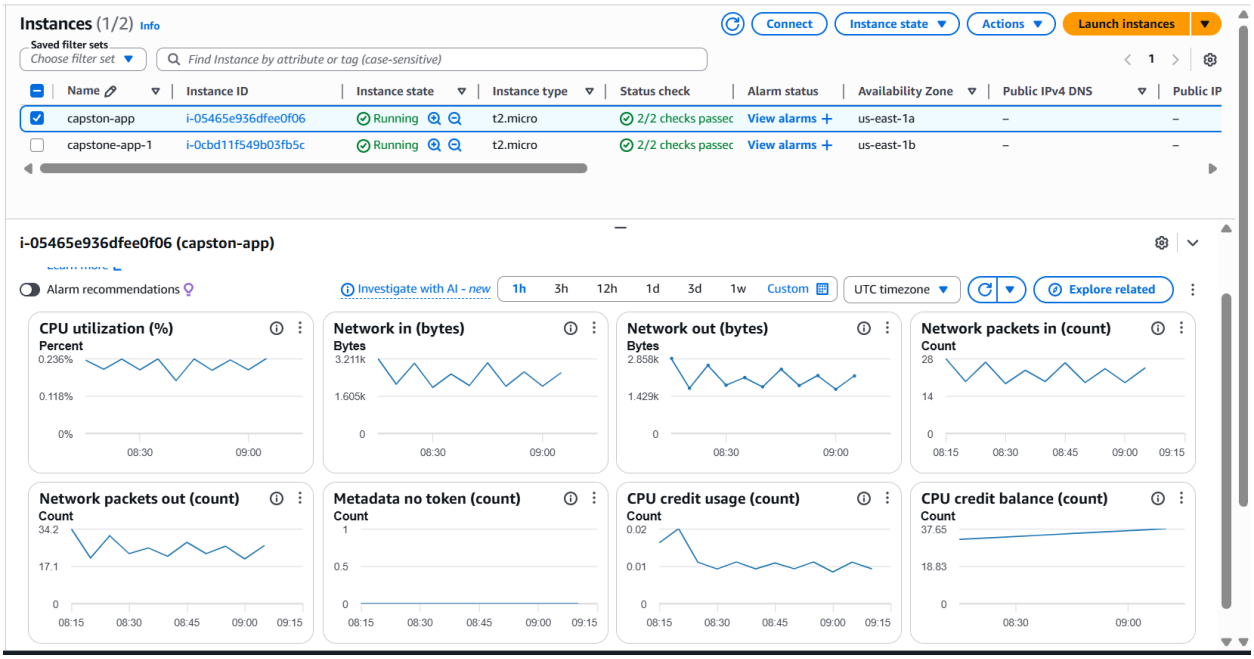
6. Security

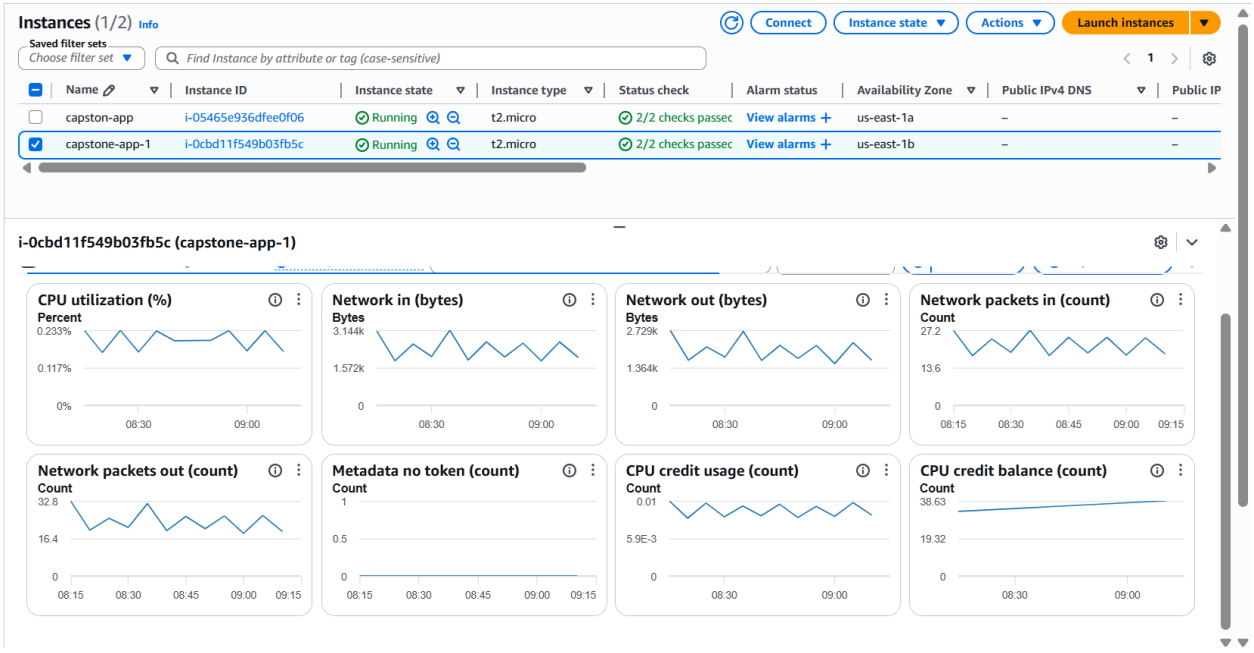


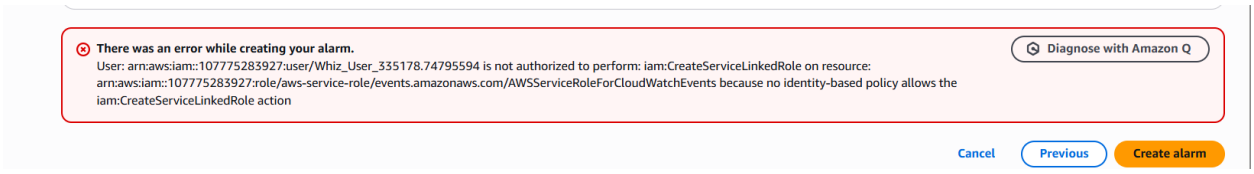




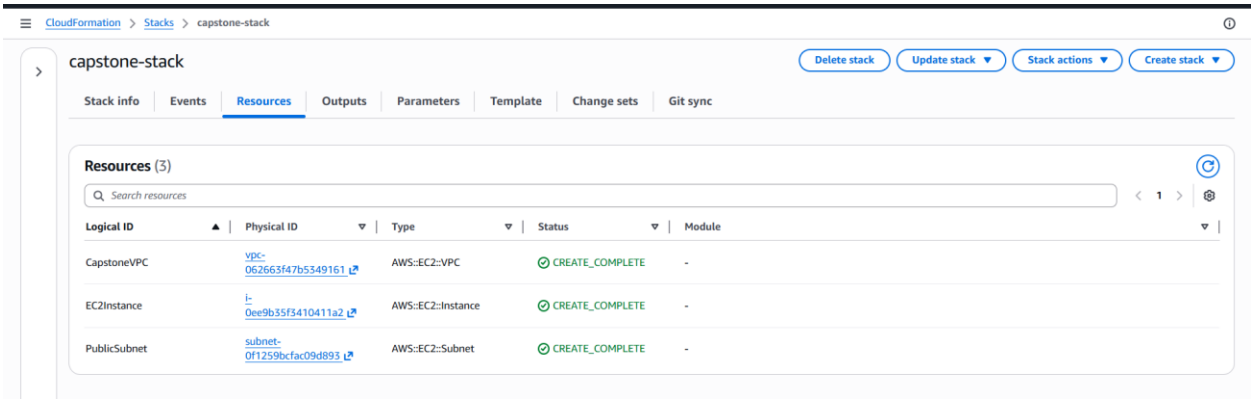
7. Monitoring & Logging

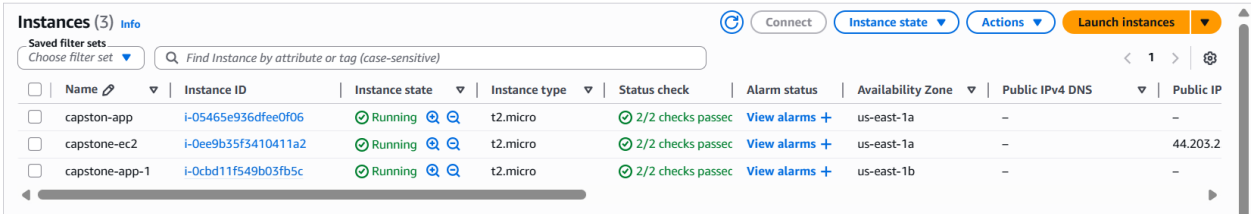


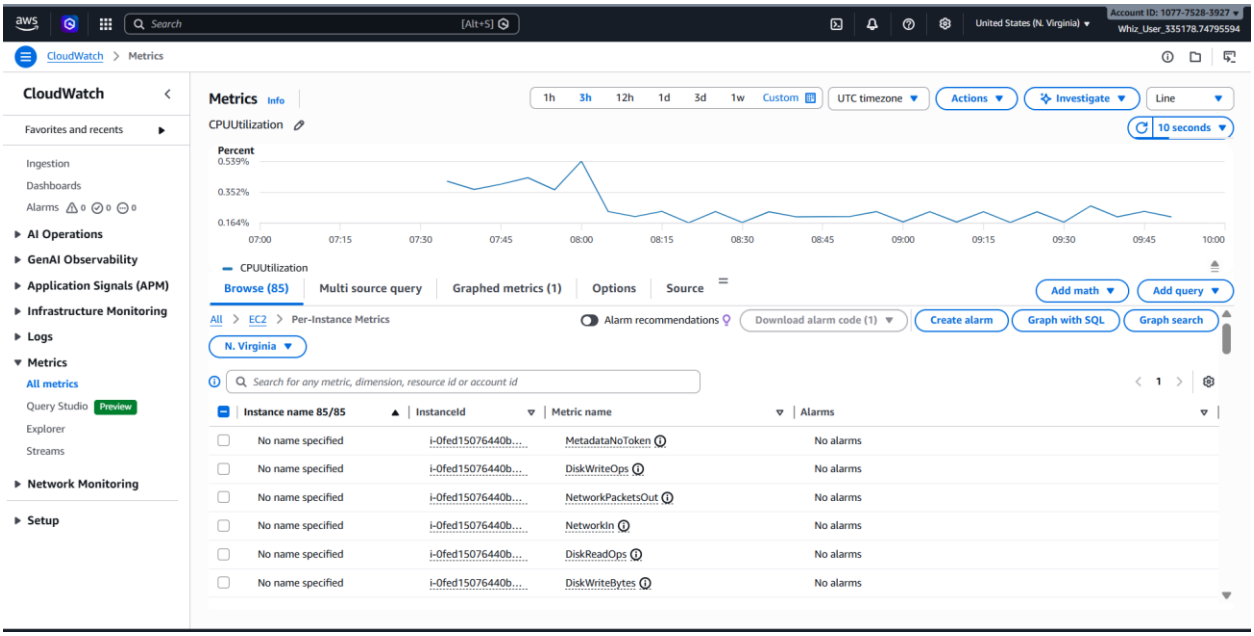




8. Automation (Mandatory)






aws
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United States (N. Virginia)
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CloudWatch > Metrics
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Favorites and recents
Ingestion
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Alarms
AI Operations
GenAI Observability
Application Signals (APM)
Infrastructure Monitoring
Logs
Metrics
All metrics
Query Studio
Preview
Explorer
Streams
Network Monitoring
Setup
Metrics
Info
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3h
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Custom
UTC timezone
Actions
Investigate
Line
CPUUtilization
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CPUUtilization
Browse (85)
Multi source query
Graphed metrics (1)
Options
Source
Add math
Add query
All > EC2 > Per-Instance Metrics
Alarm recommendations
Download alarm code (1)
Create alarm
Graph with SQL
Graph search
N. Virginia
Search for any metric, dimension, resource id or account id
Instance name 85/85
InstanceId
Metric name
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